



Association of British Insurers



Health Claims Forum

Myocardial Infarction Medical advancements and the impact on critical illness claims

Protection Technical Committee

Heart Attack review sub-group

February 2013



Agenda



- What has changed from the medical perspective
- The main claims issues
- The developments with troponin as a biochemical marker
- Interpretation of investigations that help to provide assistance in assessing severity with regards to heart attack claims
- Conclusion



The main claims issue



- There have been significant developments in recent years in the diagnosis of heart attack
 - revised diagnostic criteria for MI in 2007 (Last revision 2012)
 - developments in the use of “troponins”
- MI diagnosed but not meeting the severity level of the ABI definition – troponin criteria not being met because the newer more sensitive assays allow more accurate assessment with lower levels and being able to be taken sooner (significantly before peaking)
- Claimants are told they have suffered a “heart attack” and are expecting to be covered
- With more cases of heart attack not strictly meeting the CI definition there is a risk of inconsistency of approach between insurers
- At this point – still only affecting a small percentage of claims (est. < 5% MI claims)
- Only presents an issue for cases with ABI 2006/2011 definitions



Heart Attack definition

ABI SoBP 1999



The death of a portion of the heart muscle as a result of inadequate blood supply as evidenced by an episode of typical chest pain, new electrocardiograph changes and by the elevation of cardiac enzymes. The evidence must be consistent with the diagnosis of heart attack

- No mention of Troponin
- In the context of acute coronary syndrome, claims are likely to be met with any troponin increase above 99th %tile



Heart Attack definition

ABI SoBP 2002



The death of a portion of heart muscle, due to inadequate blood supply, that has resulted in all of the following evidence of acute myocardial infarction:

- typical chest pain;
- new characteristic electrocardiographic changes;
- the characteristic rise of cardiac enzymes, troponins or other biochemical markers;

where all of the above shows a definite acute myocardial infarction. Other acute coronary syndromes, including but not limited to angina, are not covered under this definition

- Both of the previous ABI definitions are unlikely to be affected by any changes made to the current definition
- **No severity level for troponin = in the context of acute coronary syndrome, claims are likely to be met with any troponin increase above 99th %tile**



Current ABI definition SOBP 2006 (+ 2011)



Heart attack – *of specified severity*

Death of heart muscle, due to inadequate blood supply, that has resulted in all of the following evidence of acute myocardial infarction:

- Typical clinical symptoms (for example, characteristic chest pain).
- New characteristic electrocardiographic changes.
- The characteristic rise of cardiac enzymes or Troponins recorded at the following levels or higher;
 - Troponin T > 1.0 ng/ml
 - AccuTnl > 0.5 ng/ml or equivalent threshold with other Troponin I methods.

The evidence must show a definite acute myocardial infarction.

For the above definition, the following are not covered:

- Other acute coronary syndromes including but not limited to angina.

Spectrum of acute coronary syndrome		
ACS with unstable angina	ACS with myocyte necrosis	ACS with clinical myocardial infarction
Marker: Tn and CK-MB undetectable	Troponon elevated Tn T < 1.0 ng/ml	Tn T > 1.0 ng/ml +/- CK-MB↑ or AccuTn I > 0.5 ng/ml
ECG: ST↓ or T↓ or transient ST↑ or normal		ST↑ or ST↓ or T inversion: may evolve Q waves
Risk of death*: 5–8%	8–12%	12–15%
Pathology: plaque disruption, intra-coronary thrombus, micro-emboli partial coronary occlusion ← → complete coronary occlusion		
LV function: no measurable dysfunction ← → systolic dysfunction, LV dilatation		



Universal definition of MI 2007-2012



The term myocardial infarction (MI) should be used when there is evidence of myocardial necrosis in a clinical setting consistent with myocardial ischaemia. Under these conditions any of the following criteria meets the diagnosis for MI

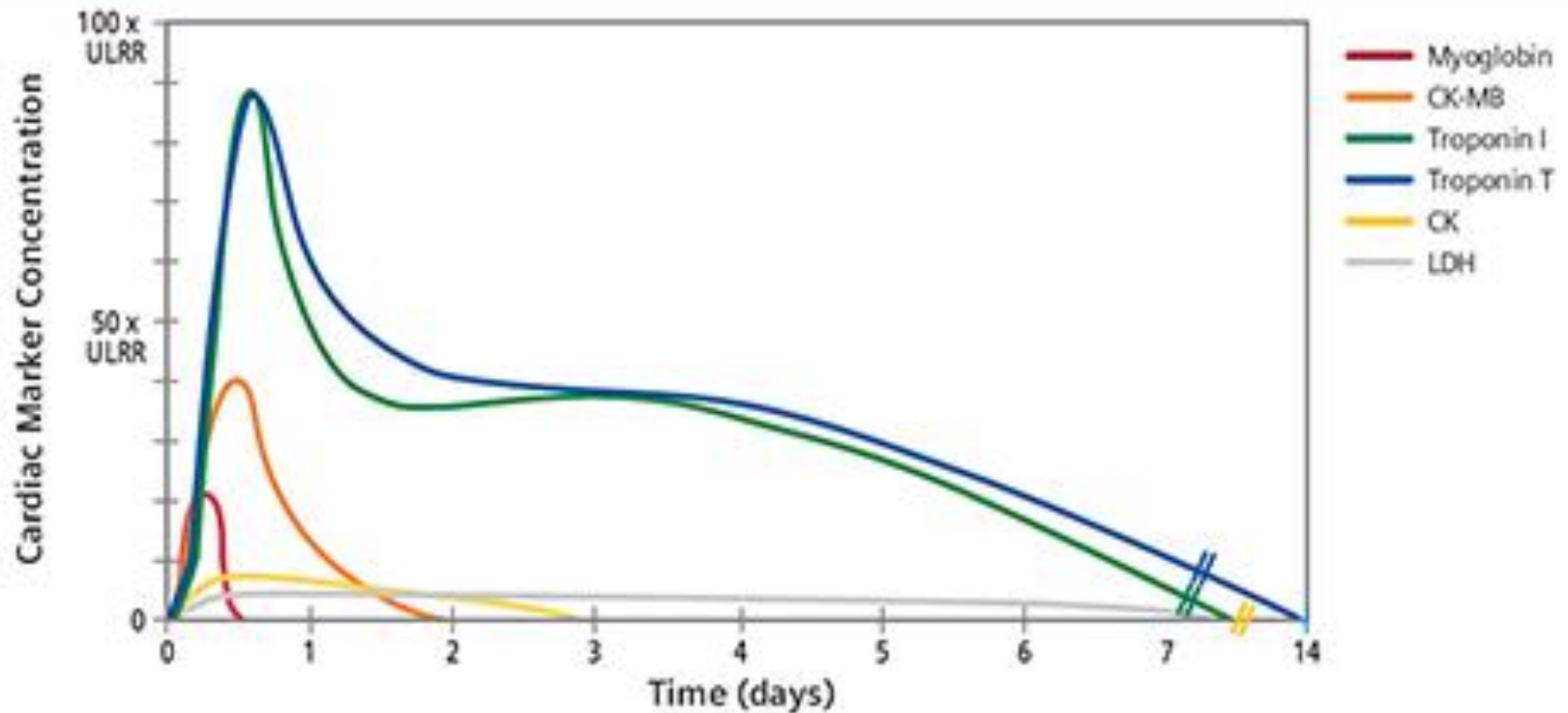
Detection of rise and/or fall of cardiac biomarkers (preferably troponin) with at least one value above the 99th percentile of the upper reference limit (URL) together with at least evidence of myocardial ischaemia with one of the following:

- Symptoms of ischaemia;
- New **or presumed new** significant ST-segment-T wave (ST-T) changes or new left bundle branch block (LBBB)];
- Development of pathological Q waves in the ECG;
- Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality.
- **Identification of an intracoronary thrombus by angiography or autopsy**

Note: Very medically complicated

Claims issue - Pattern of increase of troponins following MI

Cardiac Markers: Approximate Levels vs. Time of Onset Post MI





Troponins – Hospital example



- If the Troponin T at 6 hours is ≥ 100 ng/L this is consistent with myocardial infarction
- If the 6 hour Troponin T result is ≥ 14 ng/L but < 100 ng/L or if there are other features of cardiac ischaemia, repeat the Troponin T at 12 hours post onset of chest pain or post admission (stating the timing on the request form) to assess if it is chronic or acute cardiac ischaemia
- There are currently no consensus recommendations on the Troponin T delta change between the serial samples, however current NICE guidelines suggest a **20%*** increase between the 1st and 2nd samples to distinguish unstable angina from evolving MI. This should however be interpreted with a full clinical history and ECG findings

***Note:** This is likely to be revised to up to 100% increase (Giannitsis for Roche)



AMI cut-off points

- There are variations in approach - assumed myocardial infarction value of ≥ 50 ng/L rather than ≥ 100 ng/L after 6 hours

Method	99 th centile	AMI cut off	
		Old units	New units
Abbott AxSym	0.4 (95 th centile)	2.0	2000
Abbott ARCHITECT	0.012	0.3	300
ADVIA Centaur cTnl	0.07 (95 th centile)	1.5	150
ADVIA Centaur Tnl-Ultra	0.04	0.78	780
Access Accu Tnl	0.04	0.5	500
Dade Behring Dimension	0.07	0.6	600
Dade Behring Stratus	0.07	0.6	600
IMMULITE 2500	0.2	0.9	200
Ortho VITROS cTnl	0.08	0.4	400
Ortho VITROS Tropl		0.12	120
Tosoh AIA	0.1	0.1	100



Troponin

- Troponin T - there is only one manufacturer (Roche)
- Troponin I - 20+ assays being manufactured although not all are available for commercial use or are being used in clinical practice
- Various assays that measure troponins have different sensitivities at the lower end of the scale and so have differing 99th percentile values. However, they are testing for the same molecules i.e. troponin T or I and so, the values found at the higher levels are consistent regardless of the assay used
- More cases of acute coronary syndrome being diagnosed as MI. Possibly by 20%-30% or even higher



“High sensitivity” troponins



- Developed over the last 2 to 3 years
- “High-sensitivity” (hs) reflects the assay’s characteristics and does not refer to a difference in the form of cardiac troponin being measured
- The use of hs troponins in hospitals has spread very rapidly throughout the UK and it is only a matter of time before all hospitals are using these more sensitive types of assay
- Able to detect cellular damage to the heart muscle accurately at even lower levels and **earlier** following an episode of ischaemia than the traditional troponin assays
- Troponin being detected over the 99th percentile at much lower levels than was possible previously– being used in medical practice
- Likely to increase MI diagnosis of admissions for ACS



Troponin – Other definition concerns



- There has also been a move within the medical world to change the unit values from ng/ml to ng/L - problematical as the more zeros pass the decimal point with electronic medical records and electronic data transfer
 - Troponin T > 1000 ng/L (1.0 μ g/L or 1.0 ng/ml)
 - Troponin I > 500 ng/L (0.5 μ g/L or 0.5ng/ml)
- Claims assessors need to be careful what unit values are being applied
- Further issue – Questioning the Troponin T value
- Dr Collinson's view = Troponin T of 200ng/L is now considered more equivalent to previous CPK values used to diagnose myocardial infarction



The future for Troponins



- Troponins are unlikely to be replaced in the foreseeable future
- A recent paper published suggested the MI definition should accept a greater assay imprecision
- Co-efficient of variation (COV) of $\leq 10\%$ to COV to $\leq 20\%$ would identify more patients at risk of recurrent myocardial infarction
- This study demonstrated that such a change would have increased the diagnosis of MI by 47%
- Although Third revision of MI definition paper is not suggesting any change to COV - this outlines the danger of paying all medically diagnosed MI – no control over what changes could happen
- Highlights the danger of not maintaining a troponin level within the insurance definition



Troponin evaluation - not associated with MI

- Cardiac contusion, or other trauma including surgery, ablation, pacing, etc.
- Congestive heart failure
- Aortic dissection
- Aortic valve disease
- Hypertrophic cardiomyopathy
- Tachy- or bradyarrhythmias, or heart block
- Apical ballooning syndrome
- Rhabdomyolysis with cardiac injury
- Pulmonary embolism, severe pulmonary hypertension
- Renal failure
- Acute neurological disease, including stroke or subarachnoid haemorrhage
- Infiltrative diseases, e.g. amyloidosis, haemochromatosis, sarcoidosis, and scleroderma
- Inflammatory diseases, e.g. myocarditis or myocardial extension of endo-/pericarditis
- Drug toxicity or toxins
- Critically ill patients, especially with respiratory failure or sepsis
- Burns, especially if affecting 30% of body surface area
- Extreme exertion

Modified from Jaffe et al.,⁴ and French and White.⁵



Claims practices when evidence is not clear cut



- Troponin levels can be significantly lower than the CI definition requires and still be diagnosed as MI (with other supportive evidence) - very possible that there is no repeat taking of the troponin at a time when the troponins are likely to reach their peak
- In practice, claims assessors will look at the whole clinical picture and look very closely at all of the evidence such as symptoms, ECGs, angiography results etc.
- Careful assessment to ensure that it is a case of acute coronary syndrome rather than the troponins being elevated for another reason (as is more likely at these lower levels)
- Concern that this could lead to different claims approaches despite the same definition being used



ECGs - ST Segment Elevation



- When coupled with chest pain characteristic of ischaemia, ECGs can help not only to diagnose MI but also to provide an indication of the severity of the event
- Persistent ST segment elevation (>20 minutes) in 2 or more contiguous leads
- Blocked artery – medical emergency – often resulting in primary PCI/CABG
- STEMI – generally considered the troponin levels if taken at peak elevation would meet levels required within the definition to claim



ECG changes - Non STEMI



- Non STEMI – initial treatment focuses upon the stabilisation of symptoms – then further evaluation
- ST segment depression or T wave inversion in 2 or more contiguous leads
- Caution as these findings can also occur with other conditions
- Unlikely that troponin levels will be recorded at peak – unless repeated because of inconclusive initial troponin results (below AMI cut-off)
- Possible that Non STEMI can occur without any significant ECG changes
 - Medical definition of MI does not insist upon ECG changes



ECG changes – post event



- Pathological Q waves
 - evidence that there has been a significant cardiac event – very likely troponin levels would be met in order to successfully claim
- Other changes
 - T wave inversion
 - New Left Bundle Branch Block
- Caution as these can also occur with other conditions
 - Where these ECG changes definitely result from an ischaemic event - very likely troponin levels would be met in order to successfully claim



Other investigations



- Coronary angiography
 - confirms the presence or absence of coronary artery disease
 - useful where troponin levels are only marginally increased
- Echocardiogram
 - Measure Ejection Fraction (EF) –reduction indicates significant damage to heart muscle – if caused by MI – troponin levels would be met if taken at peak
 - Heart wall motion abnormalities e.g. hypokinesis, akinesis, dyskinesis
 - Careful evaluation required as findings can be due to ischaemia with or without MI - or can be caused by non-ischaemic heart conditions
- Radionuclide imaging
 - Can show areas of infarction and/or reversible ischaemia
- Magnetic Resonance Imaging (MRI)/X-Ray computed tomography
 - Likely to be increasingly used and will assist diagnosis



Assessment



- Read and be aware of the content of the universal MI definition paper
- Get CMOs involved for complex claims – particularly cardiologist
- Create a complete clinical picture by obtaining all diagnostic results. e.g. serial ECGs, troponin assays, medical examination, etc.
- Troponin assays
 - Note the unit of measurement for troponin recorded. i.e. ng/ml, μ g/L etc.
 - Note the reference range for the troponin assay used
 - Can be elevated for reasons other than MI
 - Where levels are low and not taken near to peak – consider other investigations that can assist in determining the severity of an ischaemic event
- Pay particular attention to ECG changes – STEMI and NSTEMI
- Take into consideration other investigations – remembering how they can assist in confirming diagnosis and severity of ischaemic events



Conclusion



- Important that the industry has a consistent approach with regards to heart attack claims where definitions are the same
- ABI feels there is no urgency to review the Heart Attack definition before the next SoBP review in 2014
 - At this point the issue we have is only affecting a very small number of claims
 - To change the definition now would require full consultation with all interested parties and would be in addition to the full review of the SoBP in less than a years time
 - Allows time to allow full consultation during 2013 with the full facts available
 - Companies can still change definitions sooner if they wish to
- The reference paper should be used and incorporated into insurers claims philosophy documents – most of this should already be included – also provides relevant information ahead of the SoBP review
- Many thanks to Dr Collinson and CMOs who have contributed



Questions

